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HSC 4933 - Python for Health Analytics

Lab 8 - Logistic and Count-Variable Regression Analysis

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Logistic Regression Analysis

1. What is the model accuracy?

The model accuracy is 0.6842, or 68.42%. This means that the logistic regression model correctly predicted about 68% of the observations in the test data.

2. Which category has the lowest precision?

Category 1 has the lowest precision. The precision for category 1 is 0.67, while the precision for category 0 is 0.69.

3. Do all categories have at least 30 supports?

No. Category 0 has a support of 25, and category 1 has a support of 13. Therefore, neither category has at least 30 observations in the test data.

4. Based on the pseudo-R-squared value, is the model good in this scenario?

The pseudo-R-squared value is 0.0625, or 6.25%. This is a relatively low value, which means the predictors explain only a small amount of the variation in the response variable. Based on the pseudo-R-squared value, this does not appear to be a very strong model for this scenario.

Count-Variable Regression Analysis

1. The model should include four significant predictors. What are they?

The four significant predictors are depression_crudeprev, selfcare_crudeprev, indeplive_crudeprev, and loneliness_crudeprev. These predictors have p-values below 0.05.

2. Which significant predictor is the most significant?

Based on the coefficient values, indeplive_crudeprev has the largest positive coefficient among the significant predictors at 0.0522. Selfcare_crudeprev has a coefficient of -0.0608, depression_crudeprev has a coefficient of 0.0137, and loneliness_crudeprev has a coefficient of 0.0113.

3. Are there any significant predictors with a VIF greater than 10?

Yes. Two of the significant predictors have VIF values greater than 10. Selfcare_crudeprev has a VIF of 30.51 and indeplive_crudeprev has a VIF of 30.95. These high VIF values suggest multicollinearity among the predictor variables.

4. How much variability is captured by the model based on the pseudo-R-squared value?

The pseudo-R-squared value is 0.04356, or approximately 4.36%. This means the model captures only a small amount of the variability in the response variable.